

Master Projects

Course on Computational Vision, A.Y. 2025-2026 Analysis of Letterforms in Medieval Painted Chapels

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General Context

These two projects are based on a dataset of images extracted from medieval painted chapels in the Alpine Arc. Each image contains a single isolated letter from inscriptions. Some of those images are well-preserved (visible, clear edges), some others are not as they have been subject to wear and tear or degradation. For a subset of chapels, the painter is known (e.g. Biazaci, Baleison, Canavesio), while for others the attribution remains uncertain.

The dataset is organized as follows:

- Each sample is an image of a single letter.
- Each sample is associated with:
 - the letter identity (e.g. a, d, e, g, h, i, s, t),
 - the chapel and possibly the location within the chapel,
 - the painter (when known) or its most likely attribution.

The objective is to use computer vision and machine/deep learning techniques to analyze stylistic variations in letterforms and relate them to painter identification.

The datasets are available at this website: [link to One-Drive](#).

Path:

- **Annotations** finales/X where $X \in \{\text{Baleison, Biazaci, Canavesio, Pietro Guido}\}$ (known authors) or $X \in \{\text{Attributions}\}$ (not-so-sure authors) or **Inconnus** for actually unknown attributions.
- (not really needed) **Dataset**, folder of raw data and extracted letters, not very well structured.

Legend for accessing the data: [Link to the Excel file](#).

Annotations are made using the VIA annotator (Oxford): [VIA annotator](#)

Project 1: Letterforms for Painter Identification

Objective

The goal of this project is to evaluate the extent to which different letters contribute to the identification of painters. In particular, the project aims to quantify the discriminative power of individual letters and to study the trade-off between frequency and informativeness.

Research Questions

- Are some letters more informative than others for identifying the painter?
- How does the frequency of a letter affect its discriminative power?
- What subset of letters yields the best classification performance?

Tasks

1. Data Exploration

- Compute the distribution of samples per letter and per painter.
- Visualize representative examples for each letter and painter.

2. Preprocessing

- Normalize images (size, grayscale, contrast).
- Optionally apply alignment or centering.
- Consider data augmentation strategies.

3. Baseline Model

- Train a classifier to predict the painter from letter images.
- Suggested approaches include CNNs or transfer learning.

4. Letter-wise Analysis

- Train separate models using only one letter type at a time.
- Evaluate performance using appropriate metrics (accuracy, F1-score).

5. Combination Experiments

- Compare performance using different subsets of letters (e.g. only **a**, only **d/s/t**, all letters).
- Investigate the effect of balancing the dataset across letters.

6. Analysis and Interpretation

- Compare the discriminative power of letters.
- Discuss the relationship between frequency and performance.
- Identify the most informative subsets of letters.

Project 2: Detection of Multiple Hands in Baleison's Works

Objective

This project investigates stylistic variation within works attributed to the painter Giovanni Baleison. Art historical evidence suggests the presence of multiple stylistic types (notably in the letter **a**), which may correspond either to different painters working for the same *atelier* or to stylistic evolution. The goal is to determine whether such variations can be identified and characterized using unsupervised learning methods.

Research Questions

- Can distinct clusters of letterforms be identified within Baleison-attributed works?
- Do these clusters correspond to different chapels, zones, or stylistic groups?

Tasks

1. Data Selection

- Focus on letters from chapels attributed to Baleison.
- Prioritize the analysis of the letter **a** (most distinctive).

2. Feature Extraction

- Extract features using either:
 - classical methods (e.g. histograms, oriented gradients...),
 - or learned representations (e.g. CNN embeddings).

3. Clustering

- Apply unsupervised clustering methods.
- Explore different numbers of clusters.

4. Visualization

- Visualize clusters and annotate them with metadata (chapel, zone).

5. Cluster Analysis

- Evaluate cluster coherence and separation.
- Investigate correlations with known stylistic or geographical groupings.

6. Interpretation

- Discuss whether the observed structure supports the hypothesis of multiple hands.
- Clearly state limitations and sources of uncertainty.